

Conventional force-aware VLA

Vision / language / state / force

Full-action prediction

task-level motion + contact adjustment

Contact changes can outpace
action updates

failure frame

Reference error

failure frame

Stale reference

failure frame

Jamming

failure frame

Nonlocal recovery

ForceDelta-VLA

Missing-force residual distillation

Frozen teacher

force / missing-force modes

Paired correction
targets

Distill

Fast residual execution

Current force / state

Reference + context

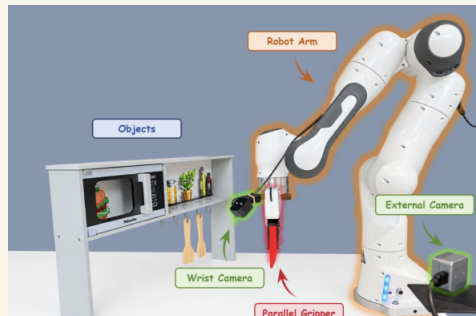
Residual policy

Missing-force
reference

Corrected action

Robot platforms

Single-arm setup



Bimanual setup

